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KEPLER-CLEAN

Automated Bayesian Pre-Whitening of A-Type Star Light Curves

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Why A-type stars?

Great transit hosts, but their pulsations get in the way.

WHAT MAKES THEM ATTRACTIVE

- Hotter, larger, more luminous than the Sun
- Wider habitable zones
- Geometrically deeper transits
- Easier to detect against Kepler's noise floor

THE COMPLICATION

- Most A-types pulsate (δ Scuti, γ Doradus, or hybrid)
- Pressure & gravity modes drive coherent oscillations
- Dozens of overlapping sinusoidal signals per star
- Can hide, mimic, or distort planetary transits

The Standard Solution

Strip the pulsations to expose the residual transit signal.

1 → 2 → 3 → 4

Compute periodogram

Lomb-Scargle on the light curve

Find dominant peak

Most powerful frequency = next target

Fit and subtract

Best-fit sinusoid at that frequency

Repeat until flat

Stop when residual looks like noise

Standard tool: Period04

Interactive GUI. A human researcher watches the periodogram update after each subtraction and decides based on intuition, when the residual looks "flat enough" to stop.

The problem with manual pre-whitening

Human stopping decisions create two failure modes, and neither is reproducible.

STOP TOO EARLY

Underfitting

- Residual pulsations still contaminate the light curve
- Real transits get buried in leftover oscillations
- False negatives or missed planets

STOP TOO LATE

Overfitting

- Noise gets fit as if it were signal
- Real transit dips can be erased
- Spurious transits can be invented

And: different analysts stop at different points, creating inter-analyst variation becomes a systematic error.

337 stars processed manually in our group so far. Scaling to thousands of Kepler A-types by hand is impractical.

Data

Kepler long-cadence light curves of A-type stars.

337	~65k	~4 yrs
A-type Kepler light curves	cadences per star (~30 min sampling)	baseline per light curve

SOURCE

Pre-processed and normalized from Kepler mission data (MAST) by Dr. Carl Melis's group at UCSD. Cervantes manually extracted frequencies for all 337 stars in Period04 which is our reference set.

DEMONSTRATION STAR FOR TODAY

KIC 4472465

52,663 cadences · 1,470.5 days · γ Dor / δ Sct hybrid candidate

Our approach

What the pipeline does between "residual light curve" and "one more component extracted."

A **Periodogram**

Lomb-Scargle on the current residual flux. Identify dominant peak ν_0.

B **MCMC**

emcee samples $p(\nu \mid \text{data})$, 16 walkers, 500 steps post burn-in, uniform prior centered on ν_0.

C **Likelihood**

Gaussian on residuals. At each candidate ν, solve A, φ, c_0 in closed form.

D **Subtract**

Best-fit sinusoid at posterior-median ν removed from light curve.

E **BIC check**

If $\Delta\text{BIC} \geq 10$ vs. previous iteration $\longrightarrow$ accept and continue. Else $\longrightarrow$ halt.

Scientific Methods

WEEK 5

MCMC

Markov Chain Monte Carlo

- Samples the posterior over frequency at each iteration
- Gives principled uncertainties, not just point estimates
- Implemented via emcee (Foreman-Mackey et al. 2013)

WEEK 6

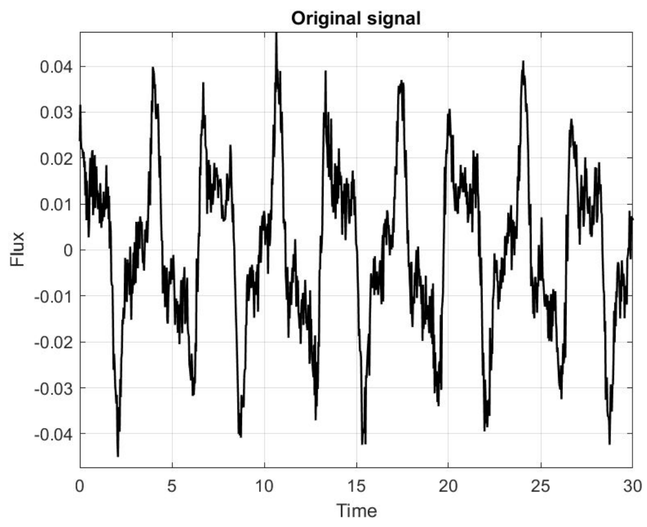
BIC

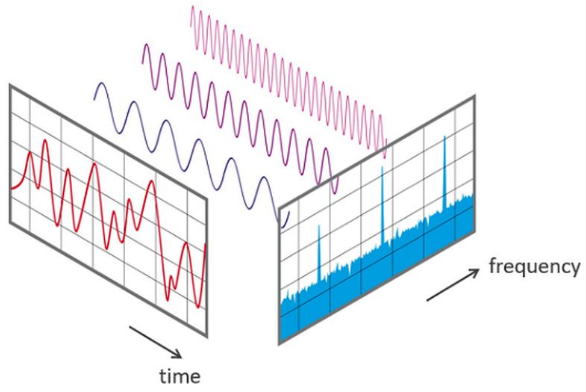
Bayesian Information Criterion

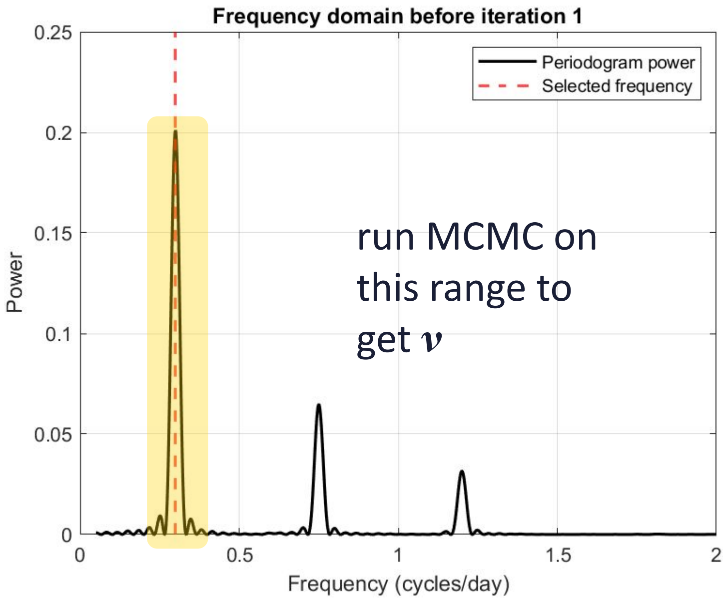
- Decides when to stop subtracting sine waves
- Each sinusoid adds 3 parameters: frequency, amplitude, and phase
- Quantitative $\Delta BIC \geq 10$ threshold (Kass & Raftery 1995)
- Replaces subjective human stopping rule

Speed trick: at fixed ν, amplitude and phase enter linearly $\rightarrow$ solved in closed form. MCMC sees a 1-D problem.

Visualize

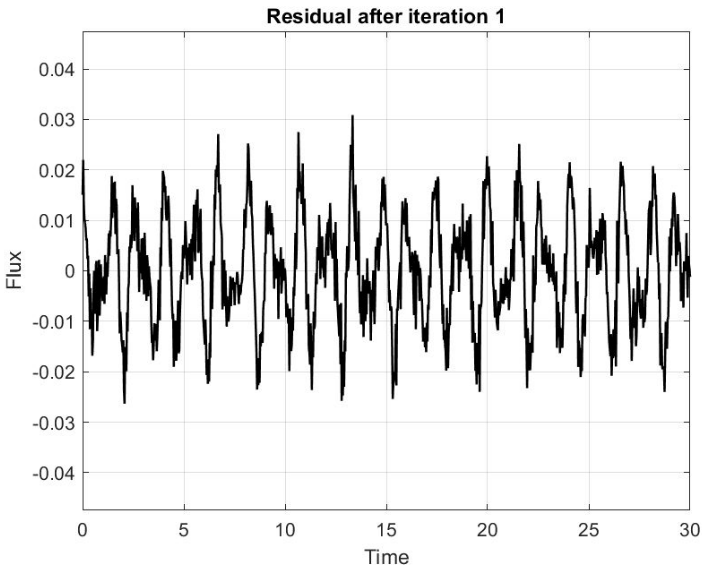






Least squares regression fit:

$$y(t) = c_1 \sin(2\pi\nu t) + c_2 \cos(2\pi\nu t)$$



Iterate Process:

$$\mathrm{BIC} = -2 \ln \mathcal{L}_0(M) + k \ln N$$

$$k = 3m$$

$$(\nu,\ c_1,\ c_2)$$

m waves subtracted and 3 parameters for each

The pipeline

Open Python package on GitHub. Modular, unit-tested.

- `prewhitening`
  - `io.py` — ASCII loader for Kepler light curves
  - `periodogram.py` — Lomb-Scargle wrapper (astropy)
  - `mcmc.py` — emcee 1-D sampler over ν
  - `sinusoid.py` — Analytic A, φ fit given ν
  - `stopping.py` — BIC and stopping rule
  - `pipeline.py` — Full iterative loop
- `batch_run.py`
- `other run files...`

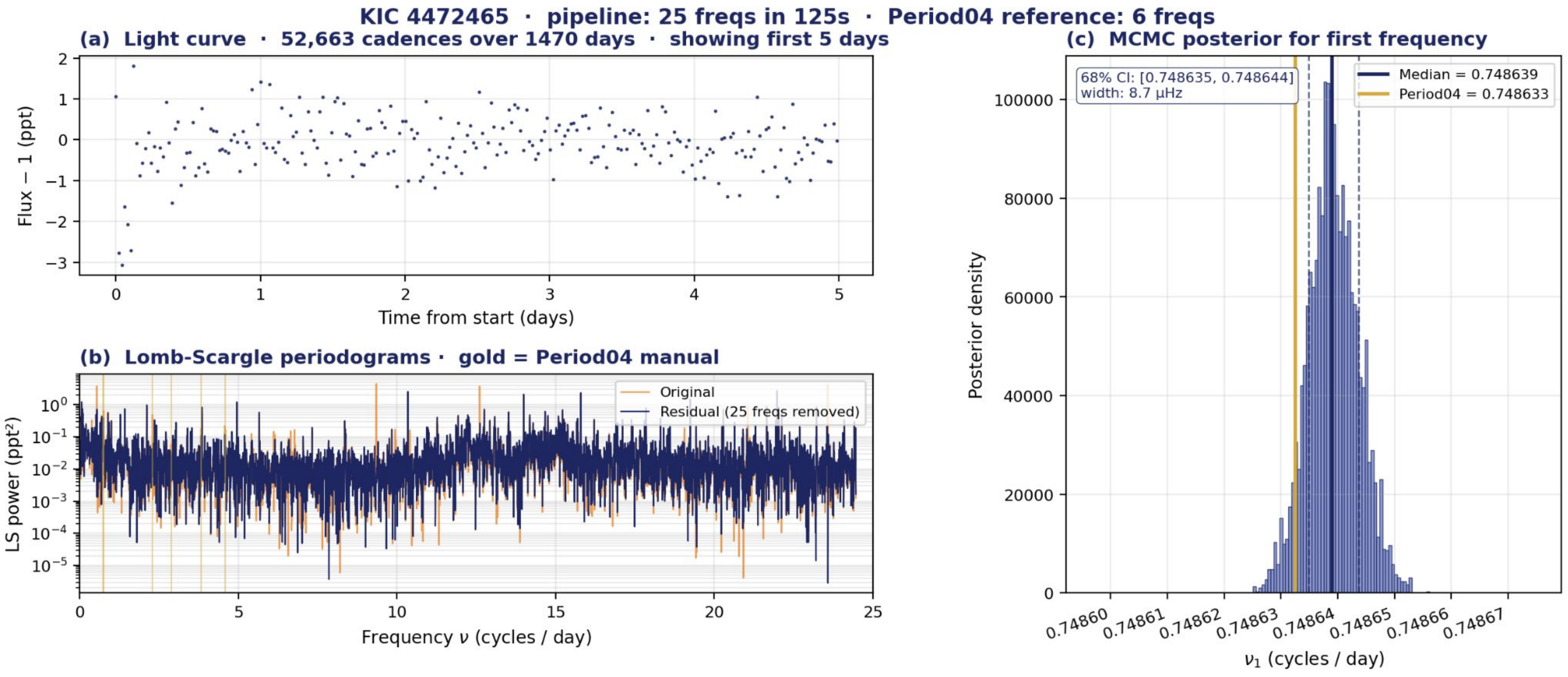
USAGE

```
from prewhitening.io import load_lightcurve
from prewhitening.pipeline import run_prewhitening

t, flux = load_lightcurve(
    "data/real/kic4472465.dat"
)
result = run_prewhitening(t, flux)

for f in result.frequencies:
    print(f.nu_median, f.amplitude)
```

github.com/kdcervantes-debug/Kepler-Clean

Results: KIC 4472465

125 s · 25 components · residual 0.505 ppt vs. Period04's 0.529 ppt

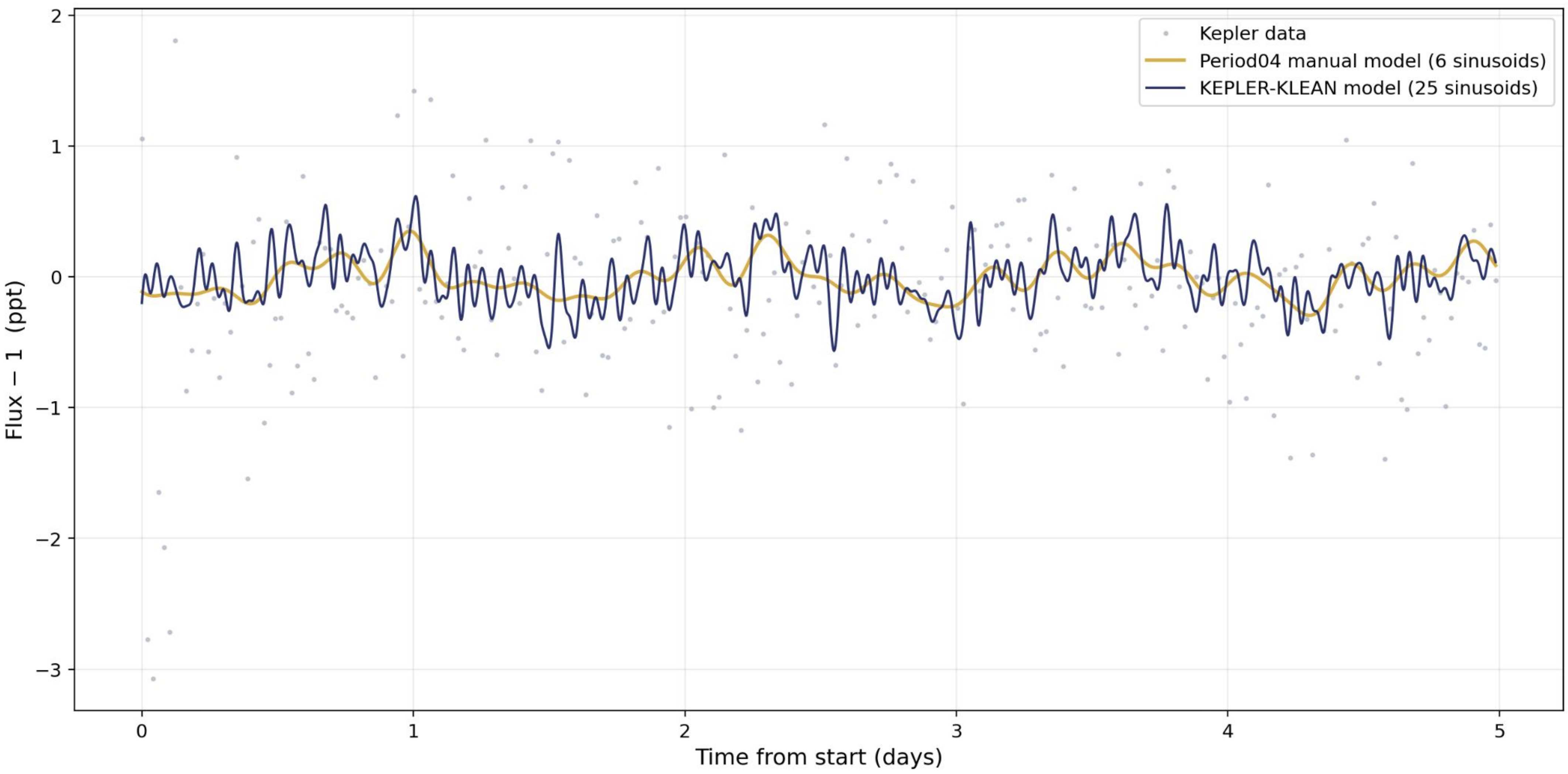


TOP EXTRACTED FREQUENCIES

#	ν (c/d)	P04 ν
1	0.748638	0.748633
2	0.728444	0.728444
3	2.289489	2.289490
4	4.586253	4.586253
5	3.823457	3.823457

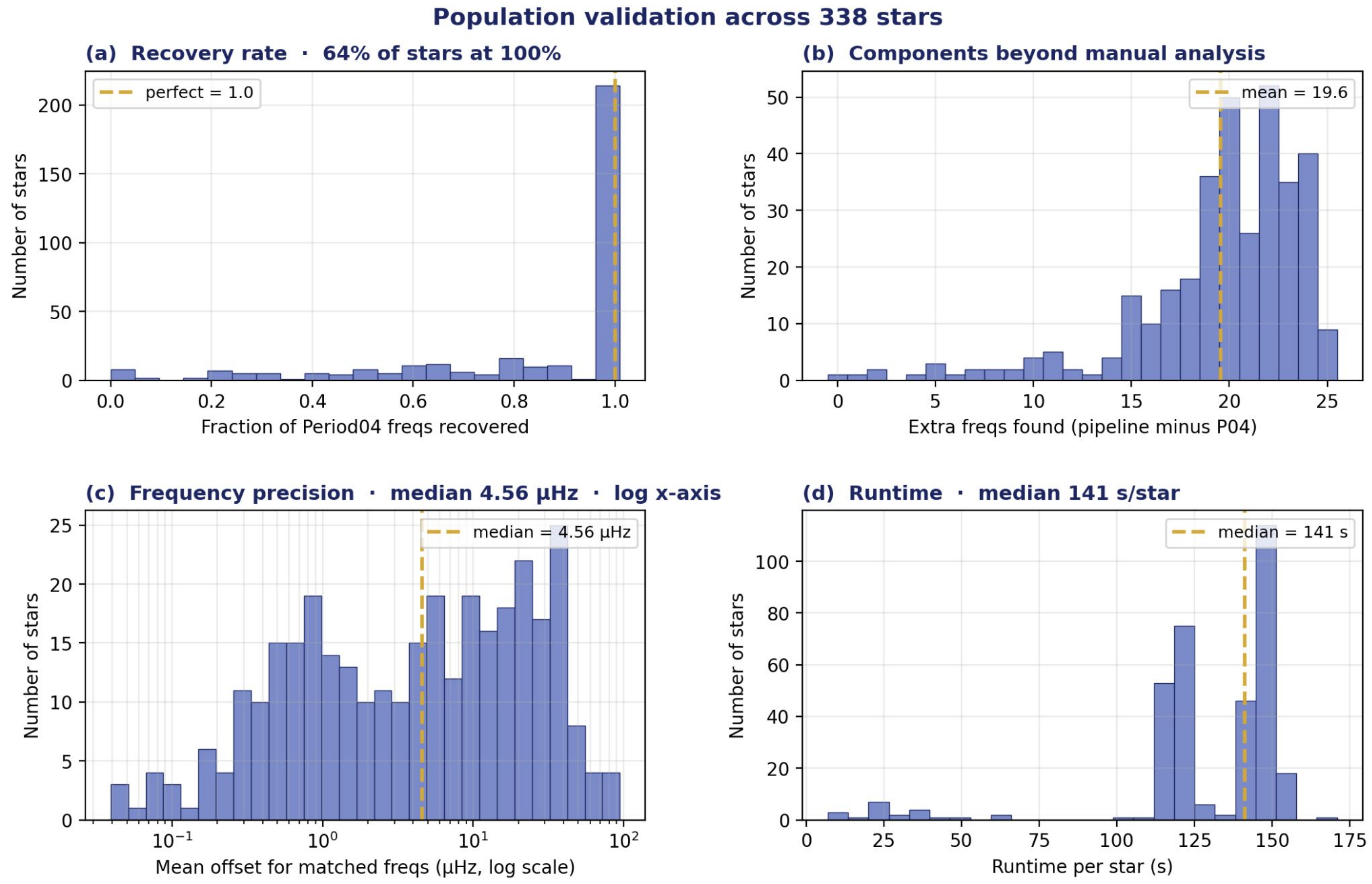
All 6 of Period04's manual frequencies recovered to within 5×10^{-6} c/d. Our pipeline kept going to 25 components; residual variance is lower.

KIC 4472465 · 6-component manual vs. 25-component automated model.



Validation across 338 stars

Median 100% Period04 recovery · mean 84% · 4.6 µHz typical offset · 141 s/star



#	Metric	Value
1	Stars compared	338
2	100% recovery	64%
3	Median offset	4.6 µHz
4	Extra freqs/star	+20
5	Runtime/star	141 s

All 338 stars processed. Tolerance 100 µHz for matching.

Median 100% Period04 recovery; 64% of stars fully recover the manual analysis. Frequency precision (~4.6 µHz) at the Kepler resolution limit. Pipeline finds ~20 additional components per star below the manual SNR threshold.

Imperfections in our Model

Assumptions baked into the current pipeline.

Correlated noise

Kepler residuals aren't 2D Gaussian, instrumental & granulation noise correlate on multiple timescales. Both BIC and our likelihood assume independence.

Convergence

Fixed step count, no autocorrelation-time check yet, some chains may be under-converged.

Noise estimator

σ from a median-filter residual MAD. Pulsation-dominated curves bias this high. Better: fit σ jointly in the MCMC.

Aliasing

30-min cadence creates aliases of high-frequency δ Sct modes. We pick the highest peak without alias checks.

Next steps

01

Inter-analyst comparison

Compare manual Period04 extractions from multiple human analysts against pipeline output to quantify inter-analyst variance and isolate where the automated method differs from human consensus.

02

Synthetic injection-recovery

Inject known sinusoids into Kepler-like noise; measure absolute frequency-recovery accuracy and false-positive rates.

03

Transit injection downstream

Inject simulated transits into pulsating curves; compare detection rates after manual vs. automated pre-whitening. Closes the loop on the original motivation.

04

Gaussian-process noise model

Replace 2D Gaussian likelihood with celerite2 GP kernels appropriate for stellar variability. Addresses the correlated-noise critique.

THANK YOU

Three takeaways

01

Automated, reproducible

MCMC + BIC removes analyst-dependent stopping decisions from pre-whitening.

02

End-to-end working

Demonstrated on real Kepler data; produces sensible extractions with posterior uncertainties.

03

Validated at scale

Demonstrated on all 338 stars: median 100% Period04 recovery at Kepler frequency resolution (~4.6 μHz). Synthetic injection-recovery and GP noise modeling ahead.

REPOSITORY

`https://github.com/kdcervantes-debug/Kepler-Clean`

Questions?

Q&A

Citations

Foreman-Mackey, D., Hogg, D.W., Lang, D., & Goodman, J. 2013, PASP, 125, 306. (emcee)

Kass, R.E. & Raftery, A.E. 1995, JASA, 90, 773. (BIC threshold)

VanderPlas, J.T. 2018, ApJS, 236, 16. (Lomb-Scargle review)

Zechmeister, M. & Kürster, M. 2009, A&A, 496, 577. (Generalised Lomb-Scargle)